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The Investment for Quantum Dot Technology for the Use of Quantum Computing with Light

TOPICS IN NANOTECHNOLOGY (CAPE1700)

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Abstract
Quantum computing is the talk of the century. Many major technology companies are seeking to exploit the benefits of it, since it harbours incomparable processing speeds and unlocks unfound capabilities in parallel processing. These companies are investing into electron quantum computing, there is however a new development concerning the use of light and its applications to develop a chip that can compute similarly to the typical quantum computer. This is known as the photonic quantum computer, and as of the date of publication, is still in development. This type of quantum computing has similar benefits to electron quantum computing, plus the low operating costs and potential of commercialisation.

As an investor, you too seek to reap the reward from this new start-up. Since the development of the technology is under control by the University of Sheffield, my offer to you is to control the basic components for this development, more specifically the quantum dot. This component is one of the core aspects of the photonic quantum chip, and as such, by controlling the supply, you can effectively control the markets and profit immensely, while also opening other opportunities into research using these chips if all goes well. Given these circumstances, this is an opportunity that cannot be avoided, for the success of the investment yields great potential for the future of both society and the company.

Contents

1 Introduction

It is conclusive that you may have heard the term “quantum computing” floating around. Being an investor, you seek to exploit the proposed, revolutionary effects in artificial intelligence (AI), financing, simulation and parallel computing applications (Stephen Gossett and Brennan Whitfield, 2022). However the traditional ideas are in some prospects flawed when considering the current state of the technology, the requirement for cooling and its scalability against the markets. This investment idea has the potential to change that, instead focusing on a new take on the technology, using light.

The fundamental technology that can power this revolutionary device lies within the field on nanotechnology, conceived by Richard Feynman in 1959 (National Nanotechnology Initiative, n.d.) concerned about the development of materials from the length scale of a nanometre. In this proposal we are more interested in the “quantum dot” and the emission of photons for quantum computation, while using reputable sources to back up claims and ideas throughout this report.

This venture will take you through the following:

- A detailed description of the quantum dot and the investment requirements for materials and machinery
- A description of the proposed method of quantum computing, including how it works, the benefits and the reasons why you would want to invest in this technology
- An analysis of the investment plans involving the quantum dot, which includes potential future investment opportunities.
- A conclusion summarising the overall outlook for this endeavour.

2 Quantum Dot Technology

2.1 What is a Quantum Dot?

Quantum dots are “tiny particles or nanocrystals of a semiconducting material with diameters in the range of two to ten nanometres (ten to fifty atoms)” (Merck, n.d.). Essentially, these crystals contain two layers or ‘bands’ in which electrons are excited and de-excited to produce photons (a particle with energy directly proportional to its frequency, which is a part of the electromagnetic spectrum (Wikipedia, n.d.)) of specific energy and wavelength. Quantum dots can be excited by either a beam of light or by inducing a current, leaving behind a hole (see figure 1).

Using figure 1, electrons from the valence band are excited to the conduction band, leaving a "hole" behind. The now excited electron de-excites, releasing a photon. The higher the band gap, the more energy required to jump between the bands and thus the greater the energy of the photon released. It is important that the photons are:

- Of the same energy and therefore wavelength, due to the difference in diffraction with different wavelengths of light.
- Emitted at the same time for quantum processing.

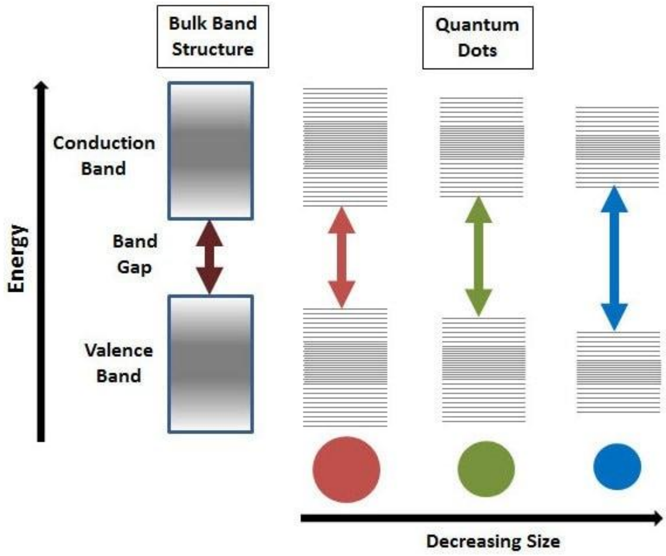

Figure 1, a depiction of a quantum dot (Merck, n.d.).

2.1.1 Photoluminescent efficiency

Photoluminescent efficiency (PLQE) is a term related to the percentage emission of outputted to inputted photons for a quantum dot (Samal et al., 2005). Ideally this would need to be maximised to make a high-quality quantum dot and allow for quantum processing, as if one dot does not produce a photon, then the computation is not possible. To maximise this, it is necessary to create a coating around the quantum dot to ensure the electron does not escape into other positive regions of the quantum crystal. Materials suitable for this must be similar in size and structure to the quantum dot crystal.

2.2 What Tools are Needed to Create Quantum Dots?

The manufacturing of quantum dots requires detailing two main aspects. These involve:

- Materials – the combinations of elements used to create the quantum dot, as well as the coating.
- Machinery – the machines and procedures needed to create the crystal and coating for the quantum dot.

2.2.1 What materials are required?

The main crystal for manufacturing quantum dots can be created from the any of the following semiconductor materials and any suitable coating material (Azonano, n.d.):

- Cadmium telluride (Merck, n.d.)
- Cadmium selenide
- Indium phosphide
- Gallium arsenide (Andrew John Vassilios Griffiths, n.d.)

The recommended material for use is Gallium arsenide, since:

- It is more widely used in research and industry today and as a consequence, has more documentation.
- It forms the structure for III-V semiconductor materials which is necessary for the use in quantum computing.

2.2.2 What machinery is required?

To construct the quantum dot to the nanoscale it is advisable to use molecular beam epitaxy (MBE). This is a method of applying single crystal thin layers to a crystal substrate under a high vacuum (Feigelson, 2015). The application of this process highly recommended, since (Andrew John Vassilios Griffiths, n.d.):

- Costs for energy can be reduced since temperatures for deposition is much lower than other methods.
- Changes can be made to the crystal due to its low and controllable growth rate.
- It allows for diagnostic techniques and sequential deposition.

The machinery used for MBE can be divided into two parts (Andrew John Vassilios Griffiths, n.d.), (Wikipedia, 2022):

- Preparation chamber – where the applied sample is injected and prepared in a Knudsen cell until sublimation.
- Growth Chamber – where the sample is 'beamed' (at an ultra-high vacuum) to a wafer, condenses and reacts with its constituent atoms (in this case, gallium and arsenic) to form a layer of the reacted substance.
    - While growing, the sample is monitored using reflection high-energy electron diffraction (RHEED), allowing for high precision application. This is vital for the development of quantum dots.

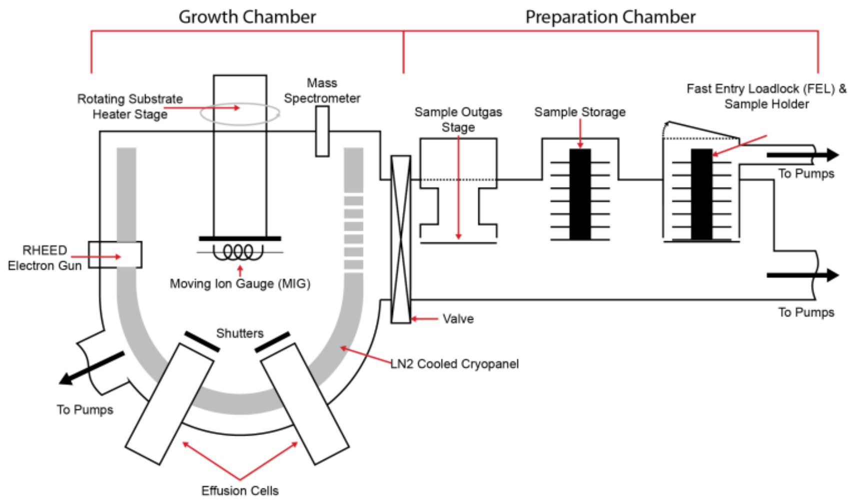

Figure 2, a depiction of the general structure of an MBE reactor (Andrew John Vassilios Griffiths, n.d.)

The creation of a quantum dot relies on the following processes (Quantum Light University of Sheffield, 2022):

- Applying two different semi-conductor materials (via MBE and using Gallium, Arsenic) with different lattices to a wafer to build up strain and creating islands (known as self-assembly).
- Applying a layer of a suitable capping material to contain these islands to improve PLQE.

3 Applying Quantum Dot Technology to Quantum Computing

3.1 What is Quantum Computing?

In order to describe the concepts behind a quantum computer, the following concepts need to be defined (Quantum Light University of Sheffield, 2015):

- Superposition – a quantum bit (or qubit) can represent the binary number zero and one at the same time. Superposition is based on the probability of these two states, and the state of superposition collapses once the qubit has been read from.
- Entanglement – connecting two or more qubits together via a non-physical interaction. This property is useful as this link allows for cause-and-effect actions, where affecting the state of one qubit can have an effect on the other(s) instantaneously.

Combining these ideas allows for faster, advanced processing of computational logic, by being able to process data without the typical architecture found in a modern computer (Quantum Light University of Sheffield, 2015).

3.2 What is Quantum Computing with Light?

Quantum computing using light or photonic quantum computing is a relatively new field of quantum computing under research. It involves the entanglement and superposition of photons (used as qubits) to compute data (Quantum Light University of Sheffield, 2015). The architecture and computational logic are based on principles derived from the Hong-Ou-Mandel effect.

3.2.1 The Hong-Ou-Mandel effect

The Hong-Ou-Mandel effect is a phenomenon that takes the identical input of photons emitted at the same time from a quantum dot and entangles the photons via a beam splitter, resulting in the following via quantum teleportation (Bouchard et al., n.d.) (Quantum Light University of Sheffield, 2014):

- A photon teleports from the left to the right
- A photon teleports from the right to the left
- Photons do not teleport
- Both photons switch sides

In practice with quantum calculations, the last two combinations are not possible since the contributions of those states are the same and cancels out. The state of the output here therefore is a superposition of the first two combinations, allowing for the basis for modelling quantum computation (Quantum Light University of Sheffield, 2014).

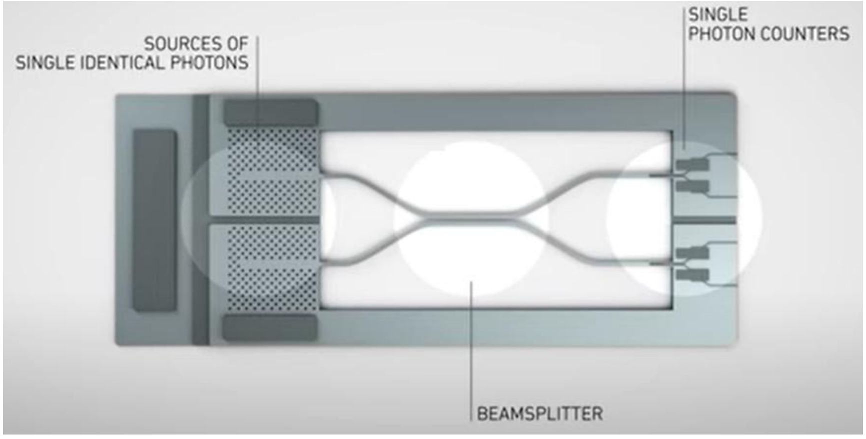

Figure 3, a basic depiction of a photonic quantum circuit. The sources of single identical photons come from shining a laser at a quantum dot (Quantum Light University of Sheffield, 2015).

3.2.2 What are the benefits of quantum computing with light?

Typically speaking, the advantages of computation with photons is as follows:

- Photons have an ability to share computation at the speed of light, making read speeds from calculations incredibly fast (Quantum Light University of Sheffield, 2015).
- The architecture derived from the Hong-Ou-Mandel effect can read the result of entangled qubits, a design that standard electron quantum computers have difficulty with (known as the measurement problem) (Slussarenko and Pryde, 2019).
- The architecture is modular and is therefore scalable, leading to exponentially increased computational power (Quantum Light University of Sheffield, 2015).
- Computation using light requires significantly less energy compared to electron quantum computing, since electrons require cryogenic freezing to absolute zero to increase computational reliability and stability (Veritasium, 2013).
- Quantum computing unlocks the ability to simulate scenarios (via the entanglement of sets of chips), providing unparalleled computational power for any system (Chertkov et al., 2022).

4 The Investment Preposition

4.1 The Investment Plan

The plan for proceeding with investment can be laid out into three phases:

- Phase one – development of all systems and preparation for small scale deployment and creation for quantum dot technology. Ideal timescale would be two to five years.
- Phase two – refinement of manufacturing time and resources, and small-scale deployment for quantum dot technology to be sold to researchers at the University of Sheffield (since they are developing the quantum chip). Potential considerations can include selling to other interested parties such as the quantum display market, or investing to expedite the development of photonic quantum computer chips.
- Phase three – large scale manufacturing for quantum dot technology for maximum profitability. Date to begin depends on the time taken for photonic quantum computers to be developed. Can be the development of new, faster methods to developing quantum dots, or planning for mass scale production. Potential new investment opportunities can involve the application of these photonic quantum chips to develop programs that exploit the benefits of parallel processing.

4.2 Costs

In order to commercialise the production of quantum dot technology, investment is required in the following (Faebian Bastiman, 2013; Faebian Bastiman, 2014):

- Initial costs to set up the facility. These include:
  - Three-phase air conditioning systems - £10,000
  - Liquid Nitrogen system - £50,000
  - Gas system - £3,000 - £5,000
  - Water system - £3,000
  - Software systems - £10,000
  - Temperature control systems - £30,000
  - A system to remove volatile organic compounds (bakeout systems) - £3,000
  - RHEED systems - £30,000
  - Remote temperature measuring systems (pyrometer systems) - £10,000
  - Growth control systems - £35,000
  - MBE costs - Typically about £1+ million for industrial grade MBE machines
- Annual costs. This includes:
  - Liquid nitrogen cooling – £12,400
  - Compressed Nitrogen or air - £1,400
  - Cell crucibles - £3000
  - Gaskets - £250
  - Cost of maintenance - £10,000
  - Substrates - £7,000
  - Cell material - £2,200 for Gallium, £5,500 for Arsenic
  - Electricity - £13,000
  - Cell coatings

4.3 Benefits for Investment

If the investment plan falls through with successful completion/continuation of phase three, the results for this endeavour are limitless. To name some (Stephen Gossett and Brennan Whitfield, 2022):

- In the short term:
  - A new facility for quantum dot manufacturing, that can potentially be retrofitted to manufacture other thin layer substances.
- In the medium term:
  - The commercialisation of a basic component for a photonic quantum chip of which, when developed, can lead to high profits.
  - The potential development of a photonic quantum chip, derived from profits of the quantum dot. If this falls through, the company becomes an industry leader in quantum computing for both business and personal use since it targets a market that constantly demands high performance with low costs.
- In the long term:
  - The investment opportunity for the applications of quantum computing. Such opportunities include:
    - Strong AI development and full manufacturing automation.
    - Battery development with new materials.
    - Finance modelling.
    - Climate change prevention and reversal schemes.
  - The overtaking of industry leaders in the development of modern computer chips for a product that functions faster for less energy cost.

5 Conclusion

In conclusion, I believe this investment opportunity has the ability to develop into a series of opportunities, each sustaining the potential to be commercialised with high profit margins. Since photonic quantum computing and its quantum dot component is relatively new and underway, this will require a hefty investment. I am however, confident that this is the direction that should be taken, considering the potential stakes in today's semiconductor manufacturing world and the plethora of benefits that comes with this technology in AI and simulation.

This is an idea built to revolutionise all ideas. I hope you can consider gathering the financing required for this ambitious endeavour.

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